



Guaranteed safe navigation via state-constraints induced by feedback control[☆]

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ABSTRACT

A fast, novel, guaranteed collision-free, sensor-based navigation technique designed using an integrated approach towards planning and control for 2D unicycle mobile robots is proposed in this work. Unlike existing works on navigation that investigate planning to initiate the design process for integration, the proposed approach commences by finding state-constraining properties of feedback control for the design of a safe navigation controller using the sensor measurements. The invariant set describing the state constraints renders an incremental planning algorithm. As a consequence, the explicit geometric structure of the state constraints is directly utilized for progressing towards the target goal point while ensuring that the system's trajectories are confined within the free regions of the workspace. The simulation and experimental results show that the proposed integrated control and planning technique not only renders a faster sensor-based navigator guaranteeing to reach the reachable target in an unknown environment but also inherently drives the robot towards maximum clearance path without explicitly planning for the same.

1. Introduction

In mobile robotic applications, it is required to enable a robot to autonomously navigate to a target point in its workspace in the presence of obstacles. Determining how to reach the target point while abiding by the space constraints imposed by the obstacles is the planning problem. The control problem involves determining the control commands to induce the required robot motion within its inherent motion constraints. To navigate to a target point while avoiding collisions, the mobile robotic system must sense the obstacles in its vicinity and compute the appropriate control commands to realize the required motion. Therefore, on a fundamental level, *moving the robot to a designated goal point in practical operating environments is the coupled control and planning problem*. Furthermore, the problem is challenging when the complete information of the environment is not available a-priori; in contrast it is perceived from the sensors mounted on the moving robot.

Given an arbitrary obstacle configuration, it is often a computationally intractable task to compute a global feedback control policy (specific to the mobile robotic system) that simultaneously abides by the free-space constraints on the workspace, and the motion constraints of the robotic system while ensuring convergence to the target point. Consequently, the chicken-and-egg problem that needs to satisfy both free-space and motion constraints has been addressed by decoupled planning and control strategies to drive the robot to a target point. In

such decoupled strategies, motion planners are employed to generate reference trajectories/paths [1,2] for safe navigation, and controllers are then independently designed for tracking [3,4].

Disturbances and the local geometry of the obstacles in the robot's vicinity are inevitable aspects to be accounted for during operation in practical environments. The following are the adverse effects that typically arise when decoupled strategies are deployed,

- The unpredictability in state transitions due to disturbances severely hinders the planning of system's state/control trajectories using optimization-based techniques for safe tracking, as frequent re-computations would be required to account for them during operation.
- If the system is thrown off course by a significant disturbance during operation, the local geometry of the obstacles needs to be accounted for while attempting to reacquire the reference trajectory to prevent any jeopardy in the system's safety. Since planning itself is a computationally intensive task, online re-planning of reference trajectories on the robotic platform would require significant resources.

To mitigate the challenges involved in the decoupled strategies, the motion planner and controller are sequentially triggered on-the-go. The local geometry of the obstacles in the robot's vicinity is accounted

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for through methods such as reactive navigation algorithms [5,6]. Vector field histograms identify safe steering directions based on the instantaneous range readings from onboard sensors [7]. The effects of disturbances have been counteracted through feedback motion planning algorithms such as artificial potential fields (APFs), navigation functions [8], harmonic potential fields (HPFs) [9], and non-holonomic RRT* [10] as they are capable of providing an updated motion plan from any new point to which the system might be dislodged to. While the potential field-based motion planners provide solutions for motion planning at the current instant, they fail to provide guarantees to work in any environment populated with obstacles for navigation. For example, they can have local minima problems due to certain obstacle configurations, and oscillatory trajectories cannot be avoided for a given parameter setting.

Consequently, navigation methods such as closest-gap navigation (CG) [11], nearness-diagram (ND) [6], and smooth-nearness diagram (SND) [12] have been developed to overcome the challenges faced by planning at an instance by increasing the planning horizon. The motion commands are typically computed based on the local sensory (range sensing) information with the primary focus being laid upon characteristics such as clearance from local obstacles during navigation and the smoothness of the resulting trajectories. The CG navigation method was shown to be more efficient than ND, and SND methods in the context of computational complexity and potential oscillations that might be induced during navigation [11]. A detailed taxonomy of collision-free navigation algorithms available in [13] conveys that the existing sensor-based local navigation strategies for the non-holonomic robot operating in an unknown workspace have their own limitations. These local navigation strategies are essential for operation in complex scenarios, as they provide real-time responsiveness, collision avoidance capabilities, and adaptability to the uncertain nature of the obstacle geometry. For example, Model Predictive Control (MPC)-based techniques are prominently state-of-the-art for addressing local navigation problems for robotic systems equipped with significant computational resources.

With the advent of sophisticated hardware, local navigation problems of underactuated, non-linear mobile robotic systems (such as wheeled mobile robots) are predominantly tackled through optimization and optimal control strategies. MPC [14] is a state-of-the-art optimization-based technique that is deployed in both the paradigms of planning amiable paths/trajectories [15], as well as for generating tracking [16] control inputs to enable local navigation in mobile robots. They are particularly adept at handling constraints as they directly encode (motion constraints due to the system dynamics, space constraints arising out of the workspace, and the input constraints from the actuators) them in their iterative, finite horizon optimization problem. Real-time MPC implementations heavily rely on linearization (and subsequent discretization) of the natural non-linear dynamics of the system for control computations over reference trajectories. They also exhibit favorable characteristics over potential field and velocity obstacle methods for the navigation of higher-order vehicle models. During real-time implementations, a balance is struck between the time horizons and the available resources so that the lag for computations is not detrimental.

Over the past decade, a rich vein of literature for their applications on mobile robots [17] has been developed. For example, MPC-based techniques are developed for designing controllers to track reference trajectories for navigation under the influence of additive disturbances in control inputs [18,19]. There also exist other robust variants such as Min-max MPC [20], Constraint tightening MPC [21], and Tube MPC [22] to deal with bounded disturbances (typically encountered in vehicle navigation problems). They are deployed in vehicle navigation problems where they iteratively carry out planning at each instant, and only the initial control input associated with the planned trajectory is applied to the vehicle. In such scenarios, it is useful as it combines planning with stability and guarantees on convergence.

However, such MPC-based approaches require significant computational capabilities as the online optimization problem that they attempt to solve often has non-convex constraints and non-linearities in the system model. Real-time demonstrations of such approaches are only carried out through approximations to ease the computational complexities [23]. Furthermore, choosing appropriate tuning parameters such as prediction horizon is vital since, short horizons might result in the controller not accounting for long-term effects, and long horizons yield sub-optimal performance in the face of variations in system dynamics while also increasing the computational complexity. Consequently, their reliability is subject to the prediction accuracy (through numerical integration of discretized system models), availability of onboard computational resources to handle online optimizations during run-time, and careful tuning of control, prediction horizons, and the weighing matrices associated with the states and the control inputs.

The generic integrated (coupled) planning and control problem seeks a motion plan in the space of admissible controls of the system for navigation. It has been addressed for the navigation problems of wheeled mobile robots [24], and balancing mobile robotic systems [25] that use known information about the static objects in the environment. The existing approaches in the literature employ the concept of sequential composition [26] to obtain a motion plan in the space of admissible controls of the system. In such approaches, control policies are defined over domains in the state space, and a structure is established such that the goal set of the control policy always lies in the domain of another control policy. Based on the instantaneous pose of the system, the planner then determines the control actions to be invoked for navigation to the target point. In the sequential composition-based approach, the control structure is pre-synthesized offline.

In summary, the existing stacks for achieving guaranteed safe navigation in unknown, cluttered environments predominantly rely on precomputing safe paths/trajectories followed by careful designs of feedback controllers for faithfully tracking them down. Computing safe trajectories in resource-constrained platforms during run-time is still a challenge as they often require solving online optimization problems based on the safety bounds sensed during run-time. Further, since the safe clearance distance constraints are non-convex in nature, their associated solutions tend to be sub-optimal in constrained environments. Safety is encoded via carefully crafting paths/trajectories with characteristics such as minimum distance to obstacles. Characteristics such as path curvature bounds are considered to ensure faithful execution within the admissible controls and kinematic constraints of the system.

Techniques such as CG, SND, ND, and their variants enable sensor-based navigation in unknown environments without the explicit requirements of pre-computed paths/trajectories. However, there are no safety guarantees since they rely on heuristics for safe control computations based on threats perceived during run-time.

Integrated planning and control approaches provide guarantees on safety for navigation by directly computing safe control inputs via the feedback controllers without the explicit requirement of solving online optimizations. This especially makes them versatile and suitable for deployment even in resource-constrained platforms. However, the existing integrated approaches in the literature are only available for safe operation in static and previously known operating environments. This brings about challenges when they must be deployed in practical environments where local sensing is often vital (to account for the local geometry of nearby obstacles) for decision-making.

Consequently, integrated strategies capable of operating online, and in real-time on the robotic platforms is a vital requirement for several practical applications where only partial information about the environment is available beforehand. Therefore, we aim to develop a safety-embedded integrated control structure in this work for guaranteed safe navigation of unicycle robots via local sensing. It accounts for the geometry of local obstacles and motion constraints (implicitly) by developing plans amiable to the onboard feedback controller during run-time.

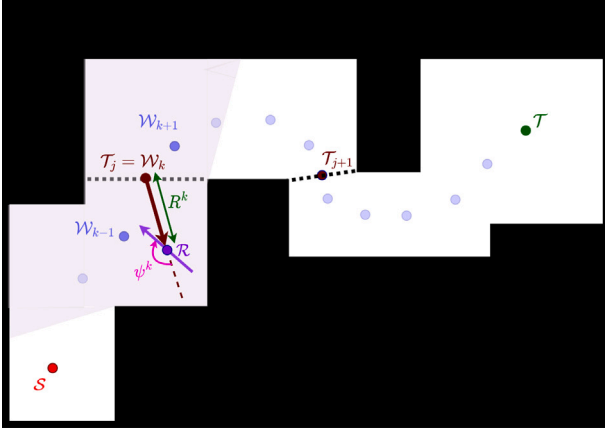


Fig. 1. An illustration to formulate the problem.

1.1. Problem formulation

An illustration in Fig. 1 shows traversable region of the operating workspace using the white shade, while the untraversable areas (obstacles) are blacked out. The purple-shaded region represents the free region of the workspace S_{free} perceived by the robot (purple dot) at $\mathcal{R}$. The brown dots indicate the intermediate target points ($\mathcal{T}_j$), where $j \in \{1, \dots, M\}$ in a set of non-zero integers. The initial problem is to assign a waypoint $\mathcal{W}_k$ (blue dots) in the perceived free region S_{free}^k at an incremental step $k \in \{1, \dots, N_{\mathcal{W}}\}$ where $N_{\mathcal{W}}$ is a non-zero positive integer. The states of the unicycle robot are its radial distance (R^k) to $\mathcal{W}_k$, and relative bearing (ψ^k) is defined around the axis that originates from $\mathcal{W}_k$ in the direction towards R^k as illustrated in Fig. 1. The control inputs are the linear (v) and angular velocities (ω). The kinematics of a unicycle robot relative to $\mathcal{W}_k$ is as follows,

$$\begin{bmatrix} \dot{R}^k \\ \dot{\psi}^k \end{bmatrix} = \begin{bmatrix} v \cos(\psi^k) \\ \omega - \frac{v}{R^k} \sin(\psi^k) \end{bmatrix} \quad (1)$$

When the robot incrementally navigates through its workspace (illustrated in Fig. 2), the perceived regions evolve with time. Each of the colored polygons in Fig. 2 represents the workspace regions visible to the robotic system when it resides at the points $\mathcal{W}_k$ at k th step. As the robot incrementally navigates through its environment, the previously occluded regions are explored until the target point $\mathcal{T}$ is perceived.

In this setting, the inter-related problems of finding invariant (safe) region with corresponding control-law (control problem), and identifying succession of invariant regions that lead to the target location (planning) that are addressed in this work are formulated as follows.

- Design of feedback control policy for the unicycle robotic system that induces an explicit invariant region in S_{free}^k with a goal set $\mathcal{W}_k$ such that, the robot's state (position) trajectories are guaranteed to evolve within this invariant region while eventually converging to $\mathcal{W}_k$.
- A sensor-based waypoint planning strategy that guides $\mathcal{W}_k$ to $\mathcal{T}_j \in S_{\text{free}}^k$ for navigation, while simultaneously ensuring overall safety (collision-free) of the system through the invariant regions (parameterized with $\mathcal{W}_k$) induced by the feedback control design.

The proposed safety-embedded integrated control structure for enabling guaranteed safe navigation is obtained by addressing the problems formulated in this subsection. The key contributions of this work that address these problems are elaborated in the subsequent subsection.

1.2. Contributions

The solution to the formulated problem is a sequence of successively overlapping invariant regions in the free workspace (constructed in real-time), induced by the feedback control policy - $U(R^k, \psi^k, \{\mathcal{W}_k\})$ for safe navigation to the target point in the workspace. From a theoretical perspective, the safe invariant region based on which the associated control inputs are computed is analytically determined. Since the feedback controller is pre-designed offline via lyapunov function-based non-linear control techniques, its associated invariant region has a well-defined geometric structure. This structure is essentially utilized to solve for the analytical solution to rapidly compute safe control inputs for navigation. The key contributions made in establishing the proposed navigator, addresses the challenges of sensor-based safe navigation in unknown constrained environments via,

- Establishing the mathematical conditions for describing invariant regions as state constraints for the feedback controller based on the spatial bounds of safety (tuples of range measurements from the onboard LiDAR) perceived during run-time.
- Designing a strategy for driving the invariant regions towards the navigation goal within the spatially varying bounds of safety perceived during run-time.

1.3. Organization

The key contributions of this work are organized as follows: Section 2 describes the proposed stabilizing feedback controller and the associated invariant region induced by the design. Section 3 describes the mathematical construction of a planner function that provides a strategy for real-time generation of waypoints $\mathcal{W}_k$ to describe the domains over the free workspace (based on the local range information) on which the feedback control policy is defined. In Section 4, the proposed navigation algorithm that dictates the control inputs to be deployed on the mobile robotic system for guaranteed safe navigation to a target point in its vicinity is presented. The proposed algorithm is experimentally demonstrated on a wheeled mobile robot and results are analyzed in Section 5.1. Working in more complex environments is verified through simulations and explained in Section 5.2. A comparative study (both qualitative and quantitative) with respect to the existing techniques for mobile robot navigation is presented in Section 6. The results show the capability of the proposed algorithm to be deployed for navigation in complex unknown operating environments through simulations, and for navigation in real-time through experimentation.

2. Control design

The navigation query consists of the starting point S , and the target point $\mathcal{T}$ to which the robot has to navigate safely. The intermediate target point ($\mathcal{T}_j$) is defined as the mid-point of the closest gap ($\mathcal{G}_j$) in the robot's vicinity. The following assumptions are considered in this work,

- The target goal point $\mathcal{T}$ is reachable, and the exact obstacle configuration in the operating workspace is unknown beforehand.
- The sensor measurements identify $\mathcal{G}_j$ (which contains $\mathcal{T}_j$) in the vicinity for navigation [11,27]. In particular, there exists a directional strategy that governs the choices in case of multiple path possibilities.

Now, the control design objectives ($\mathcal{O}_i$) of the integrated planning and control problem are as follows,

- $\mathcal{O}_1$ To orient the robot in either parallel/anti-parallel configuration to the instantaneous position vector that originates at $\mathcal{W}_k$ and terminates on the robot $\mathcal{R}$.

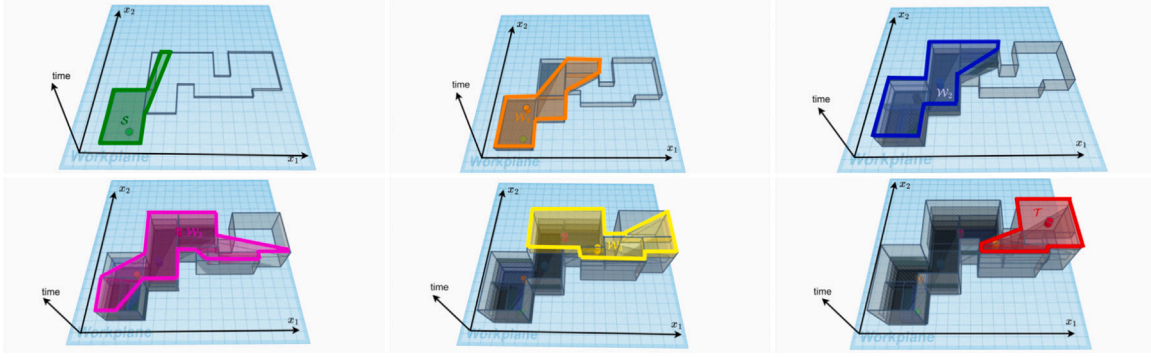


Fig. 2. The regions of the workspace perceived by the mobile robotic system at different stages during navigation.

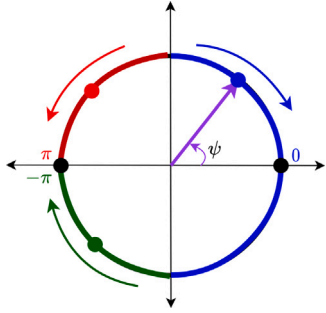


Fig. 3. Explaining the definitions of the sliding surface σ^k .

$\mathcal{O}_2$ To achieve convergence to $\mathcal{W}_k$, while simultaneously ensuring that the robot's position is constrained within an explicit region in the robot's free workspace S_{free}^k .

One of the main contributions of this work is the proposed control law presented as [Theorem 2.2](#). The control law is designed in terms of the feedback states (R^k, ψ^k) defined relative to $\mathcal{W}_k$ as illustrated in [Fig. 1](#),

$$\begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} -K_1 R^k \text{sgn}(c(\psi^k)) \\ -K_2 |\sigma^k|^{\frac{1}{2}} \text{sgn}(\sigma^k) - \int_0^t (K_3 \text{sgn}(\sigma^k) dt) \\ -K_1 \text{sgn}(c(\psi^k)) s(\psi^k) \end{bmatrix} \quad (2)$$

Here, $c(\cdot) = \cos(\cdot)$, $s(\cdot) = \sin(\cdot)$, $\text{sgn}(\cdot)$ is the signum function and K_1, K_2, K_3 are the control gains.

$$\text{sgn}(x) = \begin{cases} 1 & x > 0 \\ -1 & x = 0 \\ -1 & x < 0 \end{cases} \quad (3)$$

The definition of σ^k is based on the initial relative bearing (ψ_0^k) at the instant when it begins its navigation towards $\mathcal{W}_k$,

$$\sigma^k = \begin{cases} \psi^k - \text{sgn}(\psi^k)\pi, & -1 \leq \cos(\psi_0^k) < 0 \\ \psi^k, & 0 \leq \cos(\psi_0^k) \leq 1 \end{cases} \quad (4)$$

Lemma 2.1. The control design in (2) accomplishes $\mathcal{O}_1$.

Proof. From the definition of σ^k as in (4),

$$\begin{aligned} \dot{\sigma}^k &= \dot{\psi}^k \\ &= \omega - \frac{v}{R^k} \sin(\psi^k) \end{aligned}$$

Now, on substituting the proposed control law,

$$\dot{\sigma}^k = -K_2 |\sigma^k|^{\frac{1}{2}} \text{sgn}(\sigma^k) - \int_0^t (K_3 \text{sgn}(\sigma^k) dt) \quad (5)$$

Here, the control gains $K_2 > 0, K_3 > 0$. The kinematics in (5) is the super-twisting algorithm that guarantees desirable characteristics such as finite time convergence, robustness to disturbances whose first derivatives are bounded, and a chattering-free control. The stability and the robustness of the super-twisting algorithm has been established through linear matrix inequalities [28]. Therefore, according to the definitions of σ^k as in (4), it attains an equilibrium on the set $\{0, \pi\}$. (Considering the representation $\sigma^k \in (-\pi, \pi]$)

The definitions of σ^k as in (4) is illustrated in [Fig. 3](#). The tendency of (2) in the evolution of ψ^k to accomplish the parallel/anti-parallel configuration as stated in $\mathcal{O}_1$ is represented by the arrows. $\square$

Theorem 2.2. The control design in (2) accomplishes $\mathcal{O}_2$ while ensuring that the position trajectories are confined within an explicit invariant region L^k as stated.

Proof. From (4), the case of $\psi^k = \left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$ is not a fixed point of the system kinematics since the control design ensures that ψ^k always converges to a set of points where either of $\{0, \pi\}$ ([Lemma 2.1](#)). The stability of the proposed control design is shown using the Lyapunov function V^k as follows,

$$V^k = \frac{1}{2} (R^k)^2 \quad \text{for any } k = 1, 2, \dots, N \quad (6)$$

$$\Rightarrow \dot{V}^k = R^k \dot{R}^k = -K_1 (R^k)^2 |\cos(\psi^k)| \quad (7)$$

$$\Rightarrow \dot{V}^k = (-2K_1 |\cos(\psi^k)|) V^k; K_1 > 0 \quad (8)$$

$$\Rightarrow \dot{V}^k = -\gamma^k(t) V^k; \gamma^k(t) = 2K_1 |\cos(\psi^k(t))| \quad (9)$$

The Lyapunov function V^k in (6) is positive-definite and is defined with R^k representing the distance from the robot's position to a target waypoint $\mathcal{W}_k$. The derivative of this function in (7) is derived and shown to be negative during the entire open time interval of the maneuver to $\mathcal{W}_k$. The rate of decay $\gamma^k(t)$ of V^k in (9) has the following characteristics (from [Lemma 2.1](#)),

- $\gamma^k(t) > 0$ as $\psi^k = \left\{-\frac{\pi}{2}, \frac{\pi}{2}\right\}$ are not fixed points.
- $\gamma^k(t) \rightarrow 2K_1$ as $\psi^k \rightarrow \{0, \pi\}$, and in finite time.

Consequently, the negative definiteness of $\dot{V}^k$ during the maneuver implies that R^k monotonically decreases, ensuring stability. Therefore, the control design ensures the monotonic convergence of the position trajectories at $\mathcal{W}_k$.

The Lyapunov condition for stability also guarantees that the sub-level sets of the Lyapunov function are rendered positively invariant during stabilization. Mathematically, given a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$, its sublevel set L is defined as follows,

$$L(f, K) = \{X \in \mathbb{R}^n : f(X) \leq K\} \quad ; \quad K \in \mathbb{R}$$

Therefore, the robot's position is constrained within the following set (invariant region) during stabilization to $\mathcal{W}_k$,

$$L^k(V, V(R_0^k)) := \{R^k \in \mathbb{R}^+ \cup \{0\} : V(R^k) \leq V(R_0^k)\} \quad (10)$$

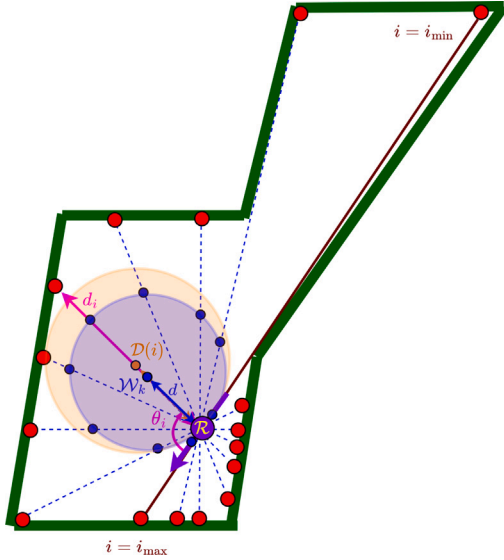


Fig. 4. A valid candidate waypoint $\mathcal{W}_k$ picked at (d, θ_i) relative to $\mathcal{R}$ with its associated invariant region L_k (blue disc). The blue dots represent the distance to the boundary points of the disc (∂L^k) along each θ_i at which d_i are obtained, and the red dots represent (d_i, θ_i) from the range sensor. The robot is indicated through the purple dot, with its instantaneous orientation represented through the purple arrow.

$$:= \{R^k \in \mathbb{R}^+ \cup \{0\} : R^k \leq R_0^k\} \quad (11)$$

From (11), the nature of the sub-level set is a disc in the robot's workspace (characterized by $L^k(V, V(R^k))$). It is also trivial to note that the robot always starts navigating from the boundary of $L^k(V, V(R^k))$ (say ∂L^k) during stabilization. (Since, $\partial L^k = \{R^k \in \mathbb{R}^+ \cup \{0\} : V(R^k) = V(R_0^k)\}$ and by definition, when the robot starts navigating $R^k = R_0^k \implies V(R^k) = V(R_0^k)$ always holds.) $\square$

3. The planner function

As the robot navigates through its environment, the local geometry of the operating workspace is perceived through onboard range sensing elements. The range information is obtained as a discrete set of points (d_i, θ_i) for each $i = 0, 1, \dots, (N-1)$, where N is the number of range measurements at the instantaneous location of the robot as illustrated in Fig. 4 (similar to the k th time step in Fig. 2). The range information provides distance (space) constraints on the angularly discretized space around the robot for safety. Therefore, for navigation to any target point in the vicinity, it is essential that the position trajectories of the robotic system abide by these constraints.

In Section 2, it was shown under the control law (2), the position trajectories of the unicycle mobile robot always evolve within an explicit invariant region in its workspace during stabilization to the target waypoint. From (11), the span of the invariant region is a function of the relative position of $\mathcal{W}_k$ at which the system stabilizes. Therefore, if $\{\mathcal{W}_k\}$ is sequentially picked such that the span of their associated invariant regions does not violate the space constraints, safe navigation to a target point is guaranteed using the proposed control design (Section 1.1).

In this section, the construction of a planner function D is proposed for defining the region in which $\{\mathcal{W}_k\}$ is chosen around the robot. (selected $\{\mathcal{W}_k\}$ ensures that the associated invariant regions are constrained within the space constraints imposed by (d_i, θ_i) .) To mathematically characterize the set of candidate waypoints $\{\mathcal{W}_k\}$, the angular region around the robot is discretized in coherence with discrete angular intervals at which the range data points are obtained.

From (11), the structure of the invariant region is a disc in the robot's workspace. Therefore, if a disc devoid of obstacles has to be

constructed at $\mathcal{W}_k$ with radius ' d' ', (The radius ' d' ' is the distance in a particular direction θ_i , for any $i = 0, 1, \dots, (N-1)$), the following conditions must hold,

$$2d \cos(|\theta_p - \theta_i|) \leq d_p \quad ; \quad \forall p \in [i_{\min}, i_{\max}] \quad (12)$$

$$i_{\min} = \text{mod}\left(i - \left\lceil \frac{N}{4} \right\rceil, N\right); \quad i_{\max} = \text{mod}\left(i + \left\lceil \frac{N}{4} \right\rceil, N\right) \quad (13)$$

Here, $\lceil \cdot \rceil$ is the ceiling function, and $\text{mod}(a, b)$ is the modulus function that returns the remainder when a/b is evaluated. In (12), it is to be noted that $|\theta_p - \theta_i|$ corresponds to the minimum angular distance between θ_p and θ_i on the unit circle, and the anti-clockwise ordering of indices is preserved in the list $[i_{\min}, i_{\max}]$. The geometric conditions in (12), (13) are illustrated in Fig. 4. The indices in $[i_{\min}, i_{\max}]$ is the angular space spanned by the disc whose diameter is along the i th index.

The planner function D is constructed such that it returns the maximum distance to the center of an obstacle-free disc at θ_i for each $i = 0, 1, 2, \dots, (N-1)$ where range data is obtained (orange shaded region in Fig. 4).

$$D(i) = \arg\max d$$

$$\text{subject to, } 2d \cos(|\theta_p - \theta_i|) \leq d_p,$$

$$d \geq 0$$

$$\forall p \in [i_{\min}, i_{\max}]$$

$$(14)$$

Since the constraints in (14) are linear, and the objective function is always monotonically increasing over the decision variable d , the planner function $D(i)$ could be equivalently computed for each $i = 0, 1, \dots, (N-1)$ as follows,

$$D(i) = \arg\min \left\{ \frac{d_p}{2 \cos(|\theta_p - \theta_i|)} \right\} ; \quad p \in [i_{\min}, i_{\max}] \quad (15)$$

Remark. Suppose $d_{\min} = \min_{p \in [i_{\min}, i_{\max}]} d_p$, then each element of $\left\{ \frac{d_p}{2 \cos(|\theta_p - \theta_i|)} \right\} \geq \frac{d_{\min}}{2}$. Therefore, a conservative solution to (15) could be simplified as $D(i) = \frac{1}{2} \min_{p \in [i_{\min}, i_{\max}]} d_p$, that corresponds to the halved minimum range reading obtained within $[i_{\min}, i_{\max}]$.

Given an angular direction for selecting $\mathcal{W}_k$, the planner function D provides a strict upper bound on the farthest point that could be selected for safe navigation in that corresponding direction (indexed by ' i '). The robot is guaranteed to safely navigate to selected $\mathcal{W}_k$ within bounds imposed by D by deploying its onboard feedback control design (Section 2).

Since D is constructed by explicitly taking into account the structure of the invariant region induced by the proposed control design for selecting navigable waypoints, it establishes a coupling between planning and control for guaranteed safe navigation. The overall feedback control policy $(U(R^k, \psi^k, \{\mathcal{W}_k\}))$ for safe navigation is incrementally obtained as a function of the robot's states defined relative to an active waypoint $\mathcal{W}_k$ such that its associated invariant region $L_k \subseteq S_{\text{free}}$.

4. The proposed navigation algorithm

In this section, the proposed algorithm to enable safe point-to-point navigation for a 2D unicycle mobile robot in static, unknown, and cluttered operating environments is presented.

4.1. Candidate waypoints

The set of candidate waypoints $(\{\mathcal{W}^c(D)\})$ around the robot is obtained based on the bounds imposed by D in the angularly discretized space at which the range data points are obtained around the robot,

$$\{\mathcal{W}^c(D)\} = \{(d, \theta_i) : (d \leq D(i), \theta_i) ; \forall i \in [0, N-1]\}$$

The waypoint for navigation ($\mathcal{W}_k$) is selected such that it is the solution to the following optimization problem.

$$\mathcal{W}_k = \min_{\mathcal{W}} \|\mathcal{W} - \mathcal{T}_j\|_2 \quad (16)$$

s.t. $\mathcal{W} \in \{\mathcal{W}^c(\mathcal{D})\}$

Here, $\mathcal{T}_j$ is the active intermediate target point computed based on the gap $\mathcal{G}_j$. The set of $\{\mathcal{W}_k\}$, $k = 1, 2, 3, \dots$ for navigation are computed from (16) based on the active $\mathcal{T}_j$. During each iteration, $\mathcal{W}_k$ progressively guides the robot towards $\mathcal{T}_j$. (Since $\mathcal{W}_k = \mathcal{T}_j$ is the unique global minimum solution of (16), and $\{\mathcal{W}^c(\mathcal{D})\}$ progressively grows closer towards $\mathcal{T}_j$ during navigation).

Remark. The waypoint selection strategy as in (16) is not restrictive, and any alternate formulations could be constructed as long as it induces incremental convergence to $\mathcal{T}_j$. Based on the results from the later section, it will be shown that this waypoint selection strategy provides enhanced safety margins.

4.2. Waypoint allocation algorithm

Algorithm 1: waypoint_allocation

Input: $\mathcal{R}, \mathcal{T}_j, \{(d_i, \theta_i)\}$
Output: $\mathcal{W}_k$
1 Compute $\mathcal{D}$ from (15)
2 Compute $\mathcal{W}_k$ from (16) using $\mathcal{D}$
3 **return** $\mathcal{W}_k$

The waypoint_allocation algorithm (Algorithm 1) generates a set of waypoints for safe navigation towards a gap/target goal point ($\mathcal{T}_j$) based on the instantaneous robot position $\mathcal{R}$.

4.3. Proposed algorithm for safe navigation

Given $(S, \mathcal{T})$ in the navigation query, the proposed Algorithm presented in Algorithm 2 provides a framework for safe navigation to $\mathcal{T}$. It computes in real-time, a feedback motion plan in the space of admissible controls of the mobile robotic system for navigation – $U(R^k, \psi^k, \{\mathcal{W}_k\})$. The proposed algorithm considers the processor's clock frequency f_s and has the tunable parameters (f_w, f_c) representing the frequency at which the waypoints and the control commands are updated.

- The lines 5–10 identify a target point ($\mathcal{T}_j$) for navigation based on the gaps in the vicinity.
- In line 13, the target reference point $\mathcal{W}_k$ (waypoint) is identified based on the waypoint allocation algorithm (Algorithm 1) for safe navigation.
- The sliding variable employed in the control design (2) is determined in lines 14–15
- In lines 17–22, the robot's states required for the feedback control design are computed relative to $\mathcal{W}_k$ to obtain the overall feedback motion plan $U(R^k, \psi^k, \{\mathcal{W}_k\})$, and is deployed on the robot. ($f_s > f_c \geq f_w$)
- The control commands are deployed (updated at the rate of f_c Hz) until the robot converges to the $\mathcal{W}_k$ or there is a timeout based on the chosen waypoint update frequency (f_w Hz). In any scenario, if the robot is dislodged from the invariant region of (2), a new waypoint for navigation is recomputed to determine the control commands for safe navigation (line 13).
- Since the control design in (2) ensures convergence to $\mathcal{W}_k$, and by construction $\mathcal{W}_k$ iteratively converges to $\mathcal{T}_j$, the proposed algorithm ensures that the robot safely navigates to $\mathcal{T}$. (When $\mathcal{T}$ falls within the line-of-sight, $\mathcal{T}_j \rightarrow \mathcal{T}$).

Algorithm 2: Proposed Navigation Algorithm

Input: $S, \mathcal{T}, f_s, f_w, f_c$
Output: $U(R^k, \psi^k, \{\mathcal{W}_k\})$
1 $\mathcal{R} \leftarrow S$
2 $(v, \omega) \leftarrow (0, 0)$
3 $k \leftarrow 1, j \leftarrow 1, \text{count_way} \leftarrow 0, \text{count_control} \leftarrow 0$
4 **while** $\mathcal{R} \neq \mathcal{T}$ **do**
 /* $\mathcal{R}$ converges to $\mathcal{T}$ ($\mathcal{T}_j \rightarrow \mathcal{T}$) */
 Read range data $\{(d_i, \theta_i)\}$
 if $\mathcal{T}$ in line-of-sight **then**
 $\mathcal{T}_j \leftarrow \mathcal{T}$
 else
 Identify $\mathcal{G}_j$ for navigation
 $\mathcal{T}_j \leftarrow$ Midpoint of $\mathcal{G}_j$
 while $\mathcal{R} \neq \mathcal{T}_j$ **do**
 /* $\mathcal{R}$ converges to $\mathcal{T}_j$ ($\mathcal{W}_k \rightarrow \mathcal{T}_j$) */
 count_way $\leftarrow 0$
 $\mathcal{W}_k =$ waypoint_allocation($\mathcal{R}, \mathcal{T}_j, \{(d_i, \theta_i)\}$)
 Re-define (R_0^k, ψ_0^k) on a modified reference frame relative to $\mathcal{W}_k$
 Define σ^k from (4) using ψ_0^k
 while $\mathcal{R} \neq \mathcal{W}_k$ **or** count_way $\leq (1/f_w)$ **and** $(\mathcal{R} \in L^k(V, V(R_0^k)))$ **do**
 /* $\mathcal{R}$ safely converges to $\mathcal{W}_k$, and (v, ω) from (2) $\Rightarrow \mathcal{R} \subseteq L^k(V, V(R_0^k))$ */
 Define (R^k, ψ^k) relative to $\mathcal{W}_k$
 Compute (v, ω) from (2)
 Append (v, ω) to $U(R^k, \psi^k, \{\mathcal{W}_k\})$
 while count_control $\leq (1/f_c)$ **do**
 /* Loop runs at frequency f_s Hz */
 Deploy (v, ω)
 count_control $\leftarrow$ count_control + $(1/f_s)$
 count_way $\leftarrow$ count_way + count_control
 count_control $\leftarrow 0$
 $k \leftarrow k + 1$
 $j \leftarrow j + 1$

Remark. The proposed algorithm relies only on the local feedback information for navigation, without an explicit requirement for global planner. Further, the control commands for navigation are computed solely based on the states defined on the ego-centric frame of the robot.

5. Results

5.1. Experimental results

Safe Navigation through waypoints generated using the proposed algorithm is experimentally demonstrated on the Turtlebot3 robotic platform (Burger Variant). It is a cylindrical structured differential drive robot with a diameter of 105 mm. Its equipped with a 360° LDS-02 laser distance sensor for sensing the local geometry of the operating workspace.

In Fig. 5, the snapshots of the experiments with the proposed algorithm at discrete time instants are displayed. The white boxes (obstacles) constrain the safe operating space around the robot. The enclosed region within the circles (red) is the invariant region L_k associated with the target waypoint $\mathcal{W}_k$ (geometric center of L_k) induced by the control design in (2). In this experimental demonstration, the robot navigates between the start (red marker) to the target goal point (green marker). The plot embedded within each of the subfigures indicates the history of control inputs applied to the robot until the current time instant. During navigation, it could be observed that the robot safely

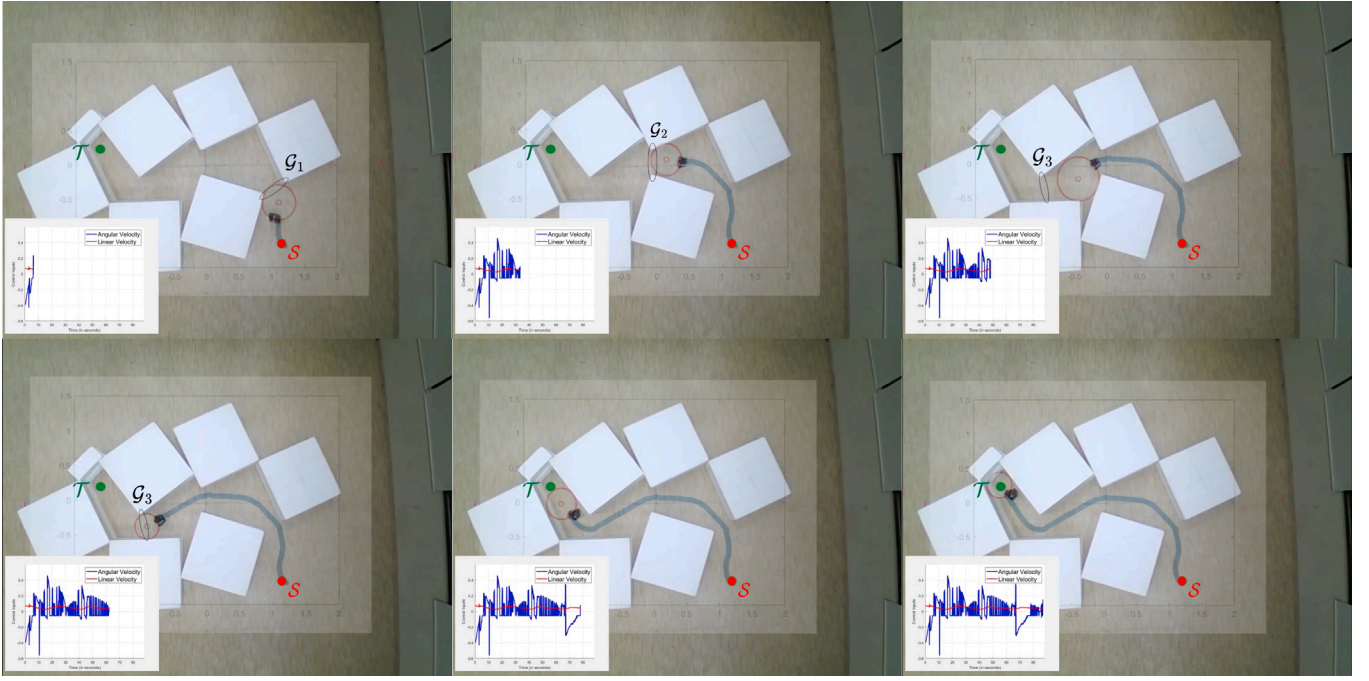


Fig. 5. The stages of Turtlebot's navigation in a chronological sequence (row-wise, left-to-right). The embedded plot shows the history of control inputs applied until that instant (Angular Velocity — Blue, Linear Velocity — Red).

Table 1

Techniques for mobile robot (unicycle) navigation in static environments — Qualitative Comparison.

S.no	Navigation method	Kinematic feasibility	Operating workspace	Computation	Planning space
1	Offline IPC [24]	Yes	Known	Offline	Controls space
2	Gap-based Navigation [6,11,12]	Yes	Unknown	Online	Workspace
3	Potential field methods [9]	No	Known	Offline (typically)	Workspace
4	Non-holonomic RRT* [10]	Yes	Known	Offline	Controls Space
5	Vector Field Histograms (VFH) [7,29]	Sometimes	Unknown	Online	Steering directions
6	Proposed Algorithm	Yes	Unknown	Online	Controls space

navigates (via invariant regions associated with the feedback control design) through the gaps G_j in the operating environment to safely reach $\mathcal{T}$.

5.2. Simulation results

In this section, simulations are carried out to demonstrate the proposed algorithm in an arbitrarily complex environment setting in MATLAB. The range readings are simulated using the LiDAR from the Navigation toolbox of MATLAB. The white regions in the simulation environment indicate the traversable regions and the blacked-out region is untraversable. The simulated LiDAR data consists of 254 range data points in a uniformly discretized angular region around the robot at each sampling instant. The boundaries to the untraversable regions in the simulated environment are purposely made non-smooth to represent any generic potential obstacle structures that could be encountered in practical operating scenarios. Further, the simulation environment has added complexity to ensure that the robot traverses through multiple gaps in a sequence to reach $\mathcal{T}$.

In Fig. 6, the stages of robot's navigation using the proposed navigation algorithm between the S (▲) and the $\mathcal{T}$ (■) are shown. The green circles indicate the boundary of the L_k (invariant regions) associated with the selected waypoints for navigation. In each of the figures in Fig. 6, the active intermediate target $\mathcal{T}_j$ and the waypoint $\mathcal{W}_k$ being pursued are highlighted. For example, in the top-left figure, $\mathcal{T}_1$ is the active intermediate target and $\mathcal{W}_4$ is the waypoint being pursued. It can be observed that the invariant regions are completely confined within

S_{free} (white regions), and the waypoints (blue ○) converge towards the active intermediate goal (black ○, computed based on the gap G_j) as the robot navigates through the workspace.

In Fig. 7, the green circles indicate the boundary of the invariant regions associated with each $\mathcal{W}_k$ generated during navigation. It is to be observed that the invariant regions are contained within S_{free} of the workspace. The red curve indicates robot's path induced during navigation.

6. Comparative analysis

6.1. Qualitative comparison

As stated earlier, navigation in cluttered environments is a combined problem of planning and control. Table 1 presents a qualitative comparison between the existing techniques employed in navigation applications for mobile robots and the proposed algorithm.

Unlike the offline IPC [24] that establishes a graph-like control structure of invariant regions over a pre-existing map of the operating workspace (with known obstacle configurations), the proposed algorithm constructs overlapping invariant regions associated with the proposed feedback control design to plan a safe motion through them during run-time. Since the proposed algorithm plans invariant regions based on the local sensory information obtained during operation, it accounts for the exact geometry of the obstacles to obtain a provably safe integrated feedback control strategy for navigation. These well-defined invariant regions are easily derived from the local range

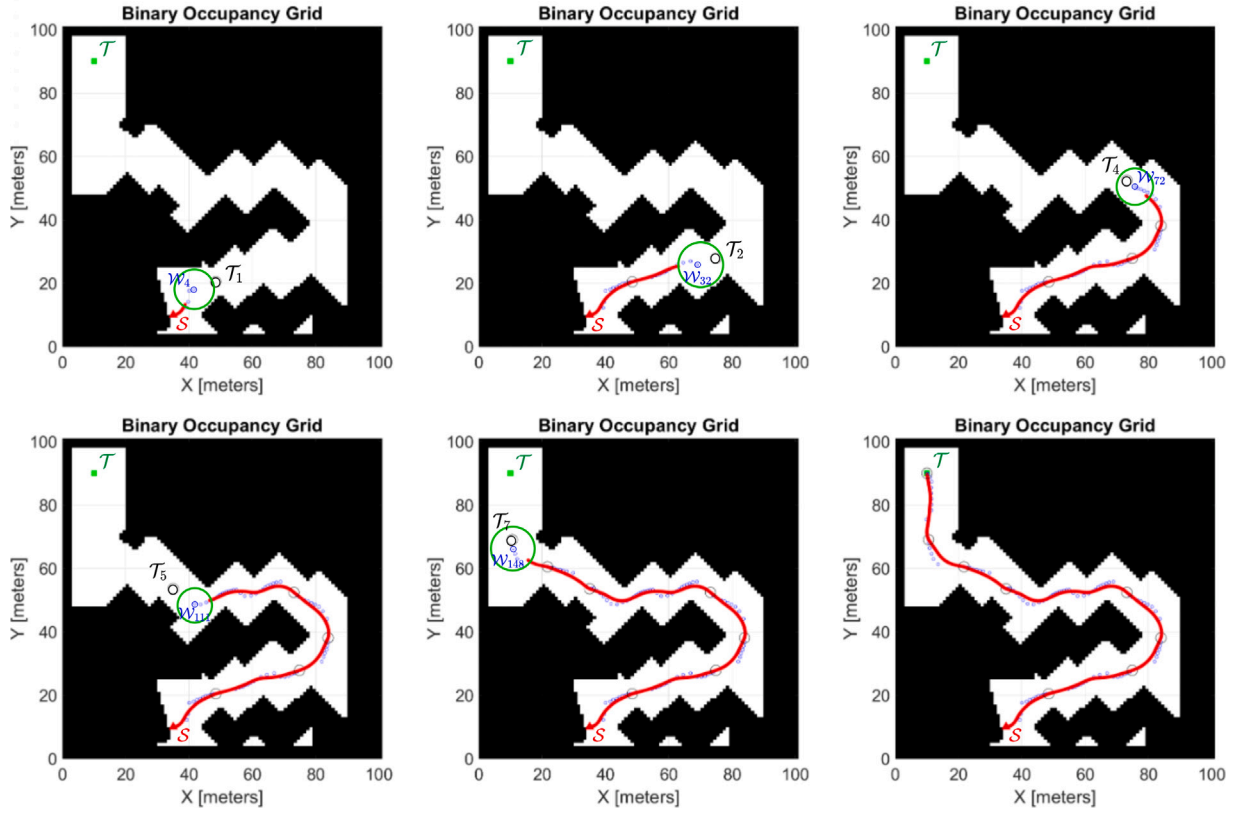


Fig. 6. Stages of robot navigation in a chronological sequence (row-wise, left-to-right) in a cluttered simulated environment.

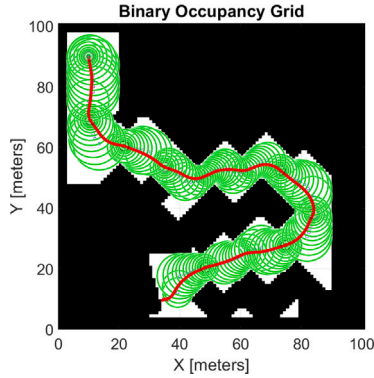


Fig. 7. Invariant Regions around the robot's path (red).

information for facilitating motion in any unknown environment. As long as the strategy of navigating through the closest G_j for seeking out $\mathcal{T}$ is successful (assumption of an available directional strategy), the proposed algorithm does not require any memory. Both the offline IPC and the proposed algorithm do not require explicit strategies for ensuring safety during navigation as they rely on the set invariance associated with the feedback control design to guarantee that the system's trajectories are confined within $\mathcal{W}_{\text{free}}$. Furthermore, unlike the offline IPC, the incremental online updates of the invariant regions in the proposed algorithm offers flexibility to deal with potential dynamic changes in the operating workspace during operation. An extension of the proposed algorithm to dynamic settings will be dealt in a future work.

In the gap-based navigation strategies [6,11,12], it is assumed that a directional strategy for navigation to $\mathcal{T}$ is available. The primary emphasis during navigation is laid upon enhancing the margins of safety,

reducing oscillations, etc. while moving through openings (gaps) in the robot's vicinity based on heuristics without any formal guarantees. The proposed algorithm also adopts a similar directional strategy, but while providing a formal guarantee of safety through the set invariance induced by the feedback control design.

As compared to the proposed algorithm, potential field [9] methods are kinematically infeasible for underactuated non-holonomic systems (unicycle robots), and they often require other reactive navigation techniques to be coupled to enhance the margins of safety while dealing with uncertainties in the exact obstacle configuration encountered during navigation.

Non-holonomic RRT* [10] is a sampling-based motion planner that provides a sequence of feedback control policies for navigation by explicitly taking into account the non-holonomic constraints in the planning phase. It requires the availability of knowledge of the workspace to a good degree of accuracy as it deals with collision avoidance by numerically verifying the induced paths during the planning phase itself. During practical deployment, it would either require the augmentation of additional reactive navigation techniques such as VFHs [7,29] to deal with the uncertainties in the workspace through sensor measurements, or a re-planning of the sequence of feedback control policies to be deployed. Since the proposed algorithm directly determines the motion commands based on the invariant regions identified during navigation, it does not require any additional augmentations during practical deployment.

6.2. Quantitative comparison

The performance of the proposed algorithm is compared with the state-of-the-art motion planning algorithm having a proportional tracking controller for navigation. We consider two cases for comparison: (1) Sensor-based planning using Artificial potential field (APF), Closest-Gap (CG) Navigation and (2) Offline planning using Rapidly Exploring Random Trees (RRT*), and Probabilistic Roadmaps (PRMs).

Table 2
Techniques for mobile robot (unicycle) navigation in static environments — Quantitative Comparison.

Metrics\Techniques→	[Sensor-based]			[Offline]		[Sensor-based]
	APF [30]	Proposed	CG [11]	PRM [31]	RRT*[31]	Proposed
	Scenario 1	Scenario 1	Scenario 2	Scenario 2	Scenario 2	Scenario 2
Computational Planning time	117.48	0.104	1.001	0.26	0.27	0.108
Average Minimum Obstacle Clearance	5.16	10.26	0.323	0.75	0.614	0.79
Path Length	75.24	77.68	8.88	9.7	10.86	9.89
Minimum obstacle clearance	4.45	9.28	0.145	0.325	0.12	0.43
Peak Curvature	0.26	0.667	∞	76.8	12.08	3.45
Average Curvature	0.025	0.039	0.94	1.33	1.12	0.66

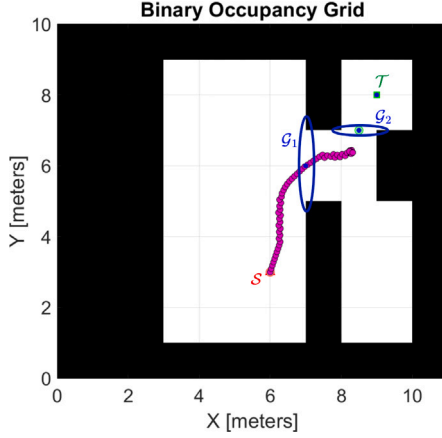


Fig. 8. Planning based on APFs.

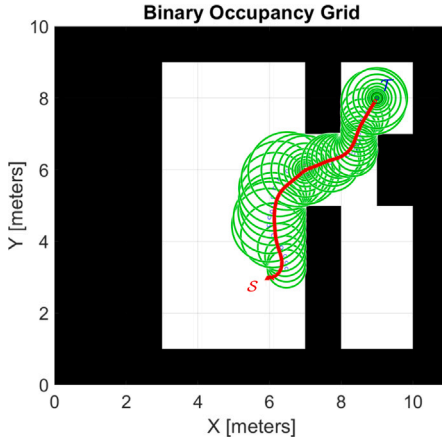


Fig. 9. Proposed algorithm. The red line indicates the path traced by the robot.

To compare with sensor-based planning, we show an example scenario in Figs. 8 and 9, where the proposed algorithm outperforms the existing APF method. To ensure fairness of comparison, sub-goals (based on the gaps) are sequentially fed to the APF algorithm for guiding the overall generation of the motion plan for navigation. As illustrated in Fig. 8, the motion plan (magenta) generated by the APF algorithm runs into the issue of local minima that results in the generated path being stuck. On the other hand, the proposed algorithm is able to accomplish the navigation objective under similar conditions.

Table 2 highlights comparison based on a set of standard metrics to demonstrate the versatility of the proposed algorithm. The quantitative comparison is further based on the following metrics:

- **Computational Time for Planning:** It is the total time taken to generate the motion plan for navigation on a computing platform. For the proposed algorithm, this planning time corresponds to the

cumulative sum of time taken to determine each of the waypoints for navigation.

- **Average Minimum Obstacle Clearance:** It is the sum average of minimum range measurement (distance to the obstacle) obtained at each sampling instant.
- **Path Length:** It corresponds to the total distance traveled by the robot while tracking the motion plan generated by the corresponding planning algorithm.
- **Minimum Obstacle Clearance:** The overall minimum obstacle clearance obtained among all sampling instances.
- **Peak Curvature:** Curvature is defined as the absolute ratio of angular velocity to linear velocity commanded during navigation. Peak curvature is the maximum value among curvatures recorded among all sampling instances.
- **Average Curvature:** It is the average value of curvatures recorded among all sampling instances.

The performance metrics are collected for sensor-based planning in a workspace resembling a part of a circular annulus as shown in Figs. 10 and 11. The dashed lines in Fig. 10 indicate the motion plan generated by the APF, while the green line shows the path traced by the robot controlled by a motion controller. The corresponding values of performance metrics are tabulated under sensor-based planning methodology in Table 2. It is evident that the proposed algorithm is around 1000 times faster than the nominal APF algorithm. The substantial improvement in saving computational time for planning is attributed to the non-requirement of the gradient computations at each planning instance in the proposed algorithm due to defining state-constraining regions. Further, the proposed algorithm induces safer motion paths when compared to the APF (the average minimum clearance to obstacles is relatively higher). However, the safer path marginally increased the path length in the case of the proposed algorithm.

Next, we compare techniques corresponding to offline planning methodology where the environment map is known apriori. In Figs. 12 and 13, sampling-based motion planners such as RRT* and PRM are illustrated in an example operating scenario. Fig. 14 illustrates obtained state-constraining regions (green circles) and traced path by the proposed algorithm in a similar setting. It is worth noting the implementation of the proposed algorithm is still sensor-based under this setting as well. The values of metrics are presented in Table 2. The proposed set invariance-based algorithm is computationally at least twice faster than both of the state-of-the-art sampling-based planning algorithms. Further, the resulting motion path traced by the proposed algorithm is relatively safer in terms of average clearance to obstacles. The path length is also comparable to the ones traced by sampling-based motion planning algorithms. In conclusion, the performance of the proposed technique is comparable to the offline planners that have full knowledge of the environment in contrast to operating only based on local knowledge gained from the on-board sensors. The proposed navigator is also shown to be superior in comparison to the CG Navigation method via all the chosen metrics except path length in Scenario 2. The slightly increased (almost comparable) path length is due to the enhanced safety margins associated with the induced path on deploying the proposed navigator.

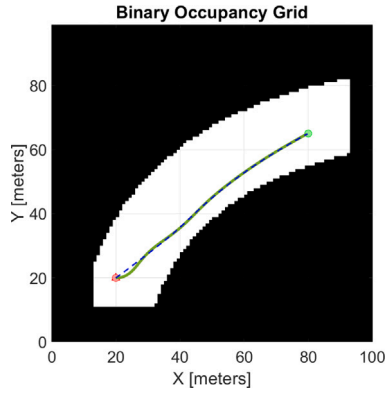


Fig. 10. (Scenario 1) APF Planning and Tracking (red marker - S , green marker - T).

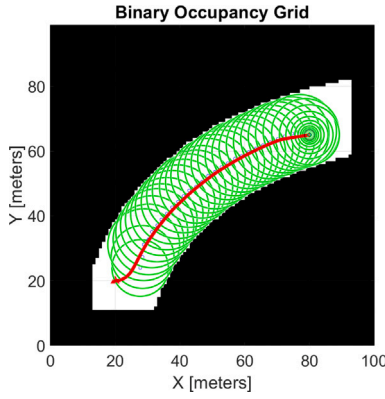


Fig. 11. (Scenario 1) Proposed Algorithm. The red line indicates the path traced by the robot.

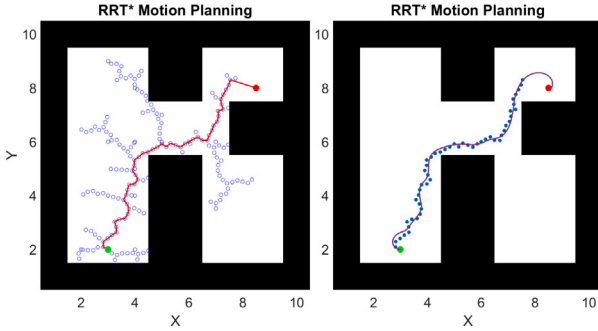


Fig. 12. (Scenario 2) RRT* Planning (left) and Tracking (right). The highlighted (magenta) portion on the left subfigure indicates the motion plan generated by the RRT* algorithm (green marker - S , red marker - T).

To present a performance comparison with model predictive control (MPC), the MPC algorithm is deployed on Scenario 2. The computational time for control computations in MPC was recorded to be 54.7 s, substantially larger than that of the proposed navigator. Also, the induced path curvatures (peak: 179.18, and average 4.105) were very large in comparison to the proposed navigator as well. This enables us to conclude that the proposed navigator in this work provides superior performance compared to MPC in these aspects (Refer Table 2 for the values of metrics of the proposed navigator).

6.3. Hardware implementation

To compare the real-time performance of the proposed algorithm, it is tested on the turtlebot3 robotic platform operating in an unknown

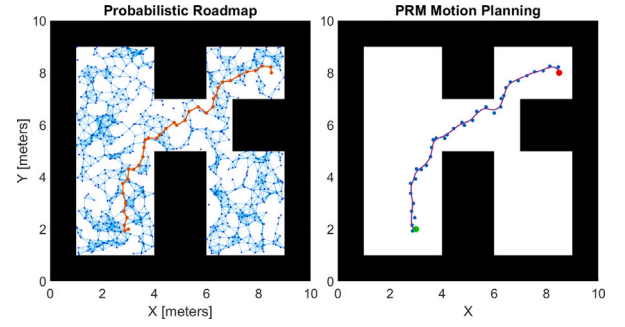


Fig. 13. (Scenario 2) PRM Planning (left) and Tracking (right). The highlighted (orange) portion on the left subfigure indicates the motion plan generated by the PRM algorithm (green marker - S , red marker - T).

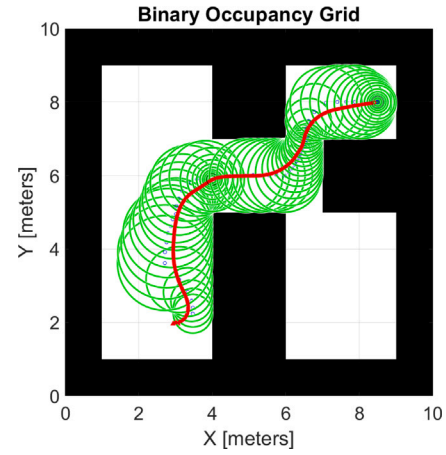


Fig. 14. (Scenario 2) Proposed Algorithm.

environment. The motion plan for navigation is computed on a laptop with a 2.3 GHz i7 processor, 16 GB RAM, and a Raspberry Pi 4 computer connected to a 32-bit ARM Cortex M7 micro-controller unit (On turtlebot3) for this comparison.

In the chosen operating scenario, the average computational time for planning individual waypoints is 12.5 ms on the laptop, while the same for onboard Raspberry Pi is around 57 ms. Due to the larger time required for motor response as compared to the reported computation time on either of the computing platform, the real-time application capability of the proposed algorithm for navigation is demonstrated.

Remark. From Table 2, the recorded curvatures (both the peak and average) on deploying the proposed method in simulations are bounded. Therefore, although the hardware implementation of the proposed algorithm is carried out on the Turtlebot3 platform, its applicability is not restricted to just differential drive platforms.

6.4. Discussion

The control input plots embedded in the experimental video snapshots show that the linear velocity profile (red) is considerably smooth, while the angular velocity profile exhibits chattering (blue). The effect of the smooth linear velocity profile results in a graceful motion convergence on the final target by pursuing the gaps sensed in its vicinity. The effect of chattering in the control input can be noticed in the experimental video when the waypoints for control computations oscillate considerably. However, the chattering is minimal when the waypoint choices are relatively smooth over successive iterations. It is also worth noting that the actuator's (motors) inherent bandwidth

limitations and the robot's inertia tend to filter the high-frequency chatters from translating onto its physical angular velocity observed during run-time. Hence, the robot's trajectory is smoother as compared to the one expected due to commanded input chattering.

7. Conclusions and future work

A fast, sensor-based navigation algorithm that operates through the set invariance guaranteed by the control design for the class of unicycle mobile robots navigating in unknown, static environments is presented in this work. The key novelty is the development of a motion plan for guaranteed safe navigation obtained during run-time through the state constraints induced by the proposed control design. Since the planning is directly associated with the feedback control design, it becomes a function of the control inputs of the unicycle robot itself. Experimental and simulation analysis on a differential drive robot (Turtlebot3) demonstrates the capability of accomplishing the navigation objectives while guaranteeing the overall safety of the system. A single sensor-based solution for control and planning enables mobile robots to accomplish guaranteed safe navigation in an unknown environment.

The potential directions of future work include the development of such strategies for complex/connected mobile robotic systems such as tractor-trailers, and extensions to similar navigation problems for flying robots in unknown 3D spaces. Further, investigating integrated control and planning for operation in dynamic environments must be a straightforward extension.

CRedit authorship contribution statement

J. Veejay Karthik: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Leena Vachhani:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: J Veejay Karthik reports financial support was provided by Prime Minister's Research Fellowship.

Code availability

The MATLAB simulation codes can be found in the github repository <https://github.com/vjkarthik97/ICPMATLAB>. The relevant ROS package consisting of the experimental implementation codes can be found at the github repository <https://github.com/vjkarthik97/ICPTurtlebot>.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.mechatronics.2024.103221>.

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